GG-C-225
JULY 1, 1958
(See sec. 6)

FEDERAL SPECIFICATION

CHEST, X-RAY FILM PROTECTIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification covers X-ray film protective chests designed to provide storage of unexposed X-ray films, or cassettes and films.

1.2 Classification.

1.2.1 *Types.*- X-ray film protective chests shall be of the following types, as specified (see 6.1):

Type I.- 1/16 inch lead lining, with four casters.
Type II.- 1/32 inch lead lining, without casters.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specifications, standards, and drawings, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

FF-C-82 - Casters, Swivel, Chair.
QQ-L-201 - Lead; Sheet.
QQ-S-636 - Steel; Carbon (Low-Carbon), Sheets and Strips.
TT-E-527 - Enamel, Synthetic, Lusterless.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

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Military Specifications:

MIL-C-132 - Crates, Open, Wood; Maximum Capacity, 2,500 Pounds.
MIL-P-6889 - Enamel, Synthetic, Lusterless.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Military Medical Supply Agency Drawings:

5012 Chest, X-Ray Film Protective 1/32 Inch Lead Lining (4 sheets).

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Satisfactory high-grade materials shall be used throughout. All parts entering into the construction of the X-ray film chest covered by this specification, shall be manufactured from materials which are entirely satisfactory for the purpose, and of sufficient strength to maintain operational alignment for life of the equipment.

3.1.1 *Steel.*- All sheet, plate, and strip stock, used in the fabrication of the items covered by this specification, and designated on Drawing 5012 as cold-rolled steel, shall conform to Federal Specification QQ-S-636 for No. 4, soft-skin-rolled temper, with dull finish.

3.1.2 *Lead.*- Lead shall conform to the requirements of Federal Specification QQ-L-201 for grade B lead.

3.2 Construction.

3.2.1 *Type I.*- Type I chest shall be so constructed that it will be X-ray proof and lightproof. All lead seams shall be soldered or burned, or shall overlap in such a way that X-rays will be excluded, regardless of position or angle of X-ray source. Chest inside dimensions shall be approximately 16-1/2 inches deep, 19 inches long, and 8 inches wide.

3.2.2 *Type II.*- All parts and assembly of type II chest shall be as shown on Drawing 5012.

3.3. Components.- There shall be four major components, as follows:

3.3.1 *Inner shell.*- Inner shell shall be made of cold-rolled, low-carbon, sheet steel as specified in 3.1.1. Sections of shell shall be welded together. Seams and joints shall have protection equivalent to that of the body material.

3.3.2 *Lead barrier.*- Inner shell shall be completely covered on ends, sides, and bottom with sheet lead, as specified in 3.1.2. The sides and ends shall be covered with a single sheet, with one vertical seam. The bottom shall be a single sheet of lead with edges folded up to overlap the joint between the bottom, sides, and ends. Lead barrier shall be completely free from pinholes. Seams and joints shall have protection equivalent to the body material.

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3.3.3 *Outer shell.*- Outer shell shall be made of cold-rolled, low-carbon, sheet steel, as specified in 3.1.1. Sections of shell shall be welded together. Seams and joints shall have protection equivalent to that of the body material specified in 3.4.

3.3.4 *Cover.*- Inner surface shall be made of cold-rolled steel covered with lead sheet. Outer surface shall be cold-rolled steel.

3.4 Detail requirements types I and II.

3.4.1 *Type I.*

3.4.1.1 *Outer shell.*- Outer shell for type I shall have four casters mounted on the bottom of the chest and the casters shall conform to the requirements of Federal Specification FF-C-82, for type I, class 2, style a, material 2 or 3. A flattop caster may be substituted, providing other requirements are met. The casters shall be so placed that when turned to the worst possible position, and with the lid open, the empty chest shall not be unstable. The casters may be welded in place, or bolted. Sheet metal or self-tapping screws of any type are not acceptable. If bolts are used, lockwashers and nuts will be used unless the thickness of the threaded stock is at least as great as the outside diameter of the bolts. Bolts used shall not be smaller than size 8-32, and there shall be at least three for each caster. Bolts and lead lining shall be so placed that the lead cannot be punctured.

3.4.2 *Cover.*

3.4.2.1 *Type I.*- Cover shall be mounted to the body of the chest by means of a substantial, continuous hinge or hinges of the 5-knuckle type, 2-1/2 inch size. The hinges may be bolted or spot-welded in position. Sheet metal or self-tapping screws of any type are not acceptable. The cover shall be held open at an angle of approximately 120° by means of a check. A drawer pull shall be welded or bolted to the cover.

3.4.2.2 *Type II.*- Cover-check mechanism shall be provided with a spring check mechanism of design as shown on drawing. Tension on springs shall be adjusted to result in a closing force of a value sufficient only to close cover. Mechanism shall also hold cover in vertical position when chest is open.

3.4.3 Thickness of materials for types I and II shall be as shown in table I.

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TABLE I.- *Thicknesses of material*

Type (see 1.2)	Inner shell (see 3.3.1)	Lead barrier (see 3.3.2)	Outer shell (see 3.3.3)	Cover (see 3.3.4)
I	24 gage (0.025 inch) or heavier, cold-rolled sheet steel.	(a) 1/16 inch thick sheet lead. (b) Single sheet of 1/16 inch thick lead.	20 gage (0.037) inch) or heavier, cold-rolled, low-carbon sheet steel.	(a) Inner surface 24 gage (0.025 inch) or heavier, cold-rolled steel with 1/16 inch thick lead sheet (b) Outer surface 20 gage (0.037 inch) cold-rolled steel.
II	20 gage (0.035 inch) cold-rolled sheet steel.	(a) 1/32 inch thick sheet lead. (b) Single sheet of 1/32 inch thick lead.	18 gage (0.049 inch) cold-rolled low-carbon sheet steel.	(a) Inner surface 20 gage (0.035 inch) cold-rolled steel with 1/32 inch thick lead sheet. (b) Outer surface 18 gage (0.049 inch) cold-rolled steel.

3.5 Finish.- All surfaces, including inside and outside exposed surfaces, and the unexposed surfaces inside the walls of the chest, shall be degreased and coated with a primer coat as specified in 3.5.1 and 3.5.2. The inside and outside exposed surfaces shall be coated with acid-, alkali-, and solvent-resisting enamel, as specified in 3.5.3.

3.5.1 *Degreasing.*- All surfaces to be enameled shall be thoroughly cleaned and chemically degreased.

3.5.2 *Primer.*- After degreasing, a primer coat shall be applied. The primer shall be zinc-chromate conforming to Military Specification MIL-P-6889. The primer coat shall form a grounding bond and base for the finished enamel.

3.5.3 *Enamel.*

3.5.3.1 *Type I.*- The primer coat shall be followed by one coat of enamel. Each coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking. The color of the final finish may be the manufacturer's standard color providing it is pleasing in appearance and suitable for the purpose.

3.5.3.2 *Type II.*- The primer coat shall be followed by one coat of lusterless enamel conforming to Federal Specification TT-E-527. Each coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking. The color of the final finish shall be lusterless green conforming to color 34087, as specified in Federal Standard 595, (see 6.3).

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3.6 Marking for identification.- The X-ray film protective chests shall be permanently marked with the manufacturer's name or registered trade-mark. Marking shall be by means of stamping or by metal plates permanently affixed.

3.6.1 *Identification marking for Military Medical Supply Agency procurement.*- In addition to the manufacturer's name or registered trade-mark, the chests shall be permanently marked with the Federal stock number, item identification, and a medical services symbol as shown on figure 1.

3.7 Welding.- All welding shall be done with appropriate equipment properly regulated and operated by thoroughly trained and experienced welders highly skilled in the operations of welding. Seam and fillet welds shall be continuous, even and smooth. Spot welds shall be not more than 2 inch centers, and shall be accomplished so as to insure sound welds of sufficient strength and completeness to tear the spot out of the parent metal when pulled apart. All exposed seam and fillet welds shall be ground smooth to eliminate traces of weld marks, and polished to blend into adjacent surfaces. All exposed spot welds shall be ground smooth, or filled and sanded smooth so that they will not be visible after the enamel finish is applied.

3.8 Workmanship.- Workmanship shall be first class throughout. The X-ray film protective chests shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.- Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL-STD-105.

4.2 Inspection.- The inspector shall ascertain that the X-ray film protective chests have been inspected and accepted in accordance with specification requirements.

4.3 Tests.- Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

4.4 Supplier responsibilities for inspection.- The supplier shall be required to perform examinations and tests to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have either adequate test facilities for executing tests, or shall make arrangements for the utilization of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be

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maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after the delivery of the supplies to which the same relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

(The levels of preservation, packaging and packing protection used are defined in Fed. Std. No. 102 (for civil agencies only)).

5.1 Packaging.- Level A.

5.1.1 *Unit of issue.*- One (1) chest, as specified, constitutes one unit.

5.1.2 *Unit preparation.*- All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.2 Packing.

5.2.1 *Level A.*- Each unit shall be overpacked in a crate constructed in accordance with Military Specification MIL-C-132. Closure and strapping shall conform to applicable paragraphs of appendix thereto.

5.2.2 *Level C.*- The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.- Shipping container shall be marked in accordance with Military Standard MIL-STD-129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.- Procurement docments should specify the following:
(a) Number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packing (see 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6525-601-2120 CHEST, X-RAY FILM PROTECTIVE, 1/16 inch Lead Lining.
6525-601-2124 CHEST, X-RAY FILM PROTECTIVE, 1/32 inch Lead Lining.

6.3 Availability of color chips.- The individual 3- by 5-inch chip of the color referenced in 3.5.3.2 is available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Price 5 cents.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 Transportation description.- Transportation description applicable to this item is:

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Chest, X-ray Film, Steel
Carload minimum weight 30,000 pounds
Chest, X-ray Film, Sheet Steel, Lead Lined
Motor Volume Minimum Weight 24,000 pounds.

6.6 Supersession data.- This specification supersedes Military Specification MIL-C-16972 *BuMed), dated 1 February 1952.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Navy - Md
Army - M
Air Force.

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